

Shouvik Baidya

Kolkata, West Bengal – 743271 — +91 9547433922 — ai.shouvikbaidya@gmail.com — LinkedIn — GitHub

Professional Summary

Aspiring Data Scientist with a strong academic background in Computer Science and Business Systems (CSBS). Proficient in Python and C, with hands-on experience in machine learning, data preprocessing, feature engineering, NLP, and time series analysis. Experienced in building, evaluating, and fine-tuning machine learning models to deliver accurate, data-driven solutions.

Education

Bachelor of Technology in Computer Science and Business Systems (CSBS) Techno International New Town	Sep 2021 – Jul 2025 CGPA: 8.33
Higher Secondary (Science) Chakdaha Ramlal Academy (WBCHE)	Jun 2019 – Jul 2021 Percentage: 84.20
Secondary (General) Chakdaha Ramlal Academy (WBBSE)	Jan 2013 – May 2019 Percentage: 90.14

Work Experience

Machine Learning Intern — Pitangent Analytics	Mar 2025 – Apr 2025
- Created sentiment analysis models using TF-IDF and Logistic Regression, achieving 77% accuracy and ROC-AUC of 0.85. - Executed feature engineering and trained XGBoost models for laptop price prediction, securing 91% accuracy. - Built and optimized a Bitcoin price forecasting model using ARIMA time series analysis, obtaining an MAE of 86.24 and an R^2 score of 0.94.	
AI Intern – Large Language Model Fine-Tuning — Pitangent Analytics	Jul 2024 – Oct 2024
- Fine-tuned TinyLlama-1.1B-Chat-v1.0 using QLoRA on a domain-specific physics question-answer dataset. - Performed data cleaning and preprocessing on 2,500+ text samples from ICSE Physics textbooks. - Improved model performance by over 20% through optimized training and loss reduction.	

Projects

Fake News Detection using NLP and Machine Learning	Jul 2025 – Aug 2025
- Designed an end-to-end NLP pipeline including text preprocessing, TF-IDF vectorization, and model training. - Evaluated nine machine learning classifiers; achieved over 99% accuracy using LightGBM on the ISOT Fake News Dataset.	
Multimodal Emotion Recognition using Deep Learning	Dec 2024 – May 2025
- Engineered multimodal models integrating text, audio, and video using Whisper, Wav2Vec 2.0, FER2013, and RoBERTa. - Applied Explainable AI techniques to improve interpretability of emotion classification results.	
Explainable AI-Based Crop Recommendation System	Jul 2024 – Nov 2024
- Developed machine learning models using agricultural, soil, and climate datasets. - Implemented SHAP and LIME for explainable and transparent AI-driven recommendations.	

Skills

- Programming Languages:** Python, C
- Machine Learning:** Supervised Learning, Classification, Regression, Feature Engineering, Model Evaluation
- Data Science:** Data Preprocessing, Data Cleaning, Exploratory Data Analysis, Time Series Analysis
- NLP & AI:** TF-IDF, LLM, Fine-tuning, Hugging Face Transformers, QLoRA, Explainable AI
- Tools & Frameworks:** Whisper, Wav2Vec 2.0, Xception, RoBERTa

Publications

Anindya Sundar Layek, P. K. Mondal, **S. Baidya**, et al., “*Explainable AI-based Crop Recommendation Systems for Improving Yield Predictions*”, 1st International Conference on Sustainable Communication, Machine Intelligence, and Metaverse, SCMIM 2025. doi.org/10.1007/978-981-96-9650-5_42

Certifications

- CISCO Networking Academy – Introduction to Data Science** — Credential
Gained a foundational understanding of Data Science concepts, lifecycle, and real-world applications, with emphasis on data-driven analytics and engineering use cases.
- Oracle Cloud Infrastructure Data Science Professional** — Credential
Strengthened proficiency in building, training, and deploying scalable machine learning models using Oracle Cloud Infrastructure and industry-standard Data Science tools.
- Oracle Generative AI Professional** — Credential
Acquired Validated expertise in designing, deploying, and managing generative AI solutions, including model architecture, prompt engineering, and ethical AI considerations.